



EDU TERIA

72nd BPSC

Prelims Online Batch

BIOLOGY

Topic wise
Notes

LECTURE

28

Plant Physiology



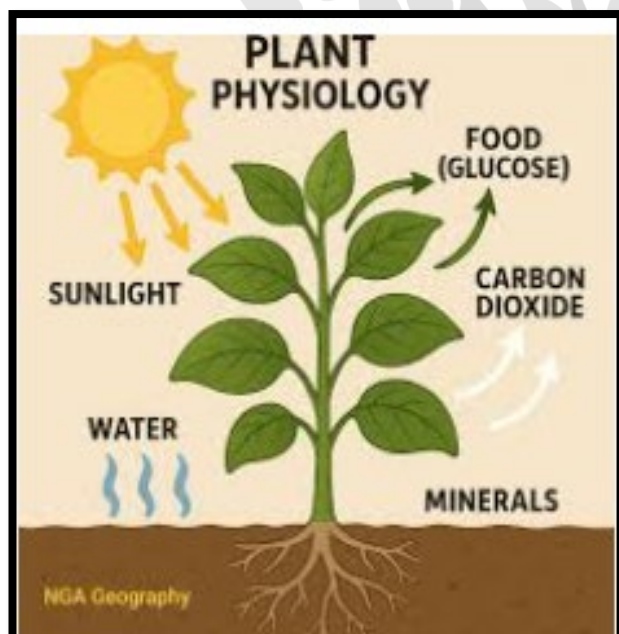
Plant Physiology

Plant Physiology

Plant physiology is the study of how plants grow, develop, and function, encompassing photosynthesis, respiration, water transport, and environmental responses. In short, it studies how Plant lives and work.

Plant Physiology consists of following things

- Photosynthesis
- Plant Hormone
- Plant Reproduction
- Plant mineral and nutrition
- Plant Respiration



Class	Function(s)	Practical uses
Auxins	Shoot elongation	Thin tree fruit, increase rooting and flower formation
Gibberellins	Stimulate cell division and elongation	Increase stalk length, increase flower and fruit size
Cytokinins	Stimulate cell division	Prolong storage life of flowers and vegetables and stimulate bud initiation and root growth
Ethylene generators	Ripening	Induce uniform ripening in fruit and vegetables
Growth inhibitors	Stops growth	Promote flower production by shortening internodes
Growth retardants	Slows growth	Retard tobacco sucker growth

Plant Reproduction

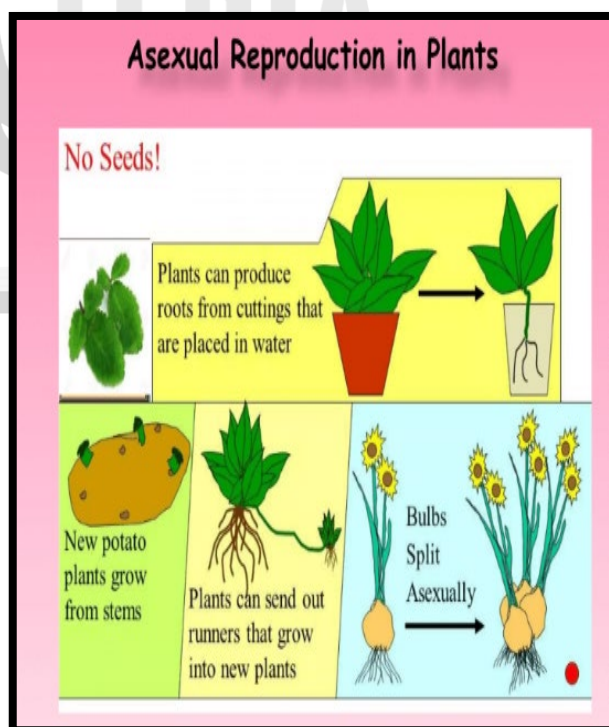
- **Plant reproduction occurs in two main ways:** Asexual reproduction, which produces genetic clones of a single parent, and sexual reproduction, which involves the fusion of gametes to create genetically diverse offspring.

Asexual Reproduction: -

- Asexual reproduction involves only one parent and does not require seeds or spores for some methods.
- The resulting new plants are genetically identical to the parent plant.

Plant Hormone

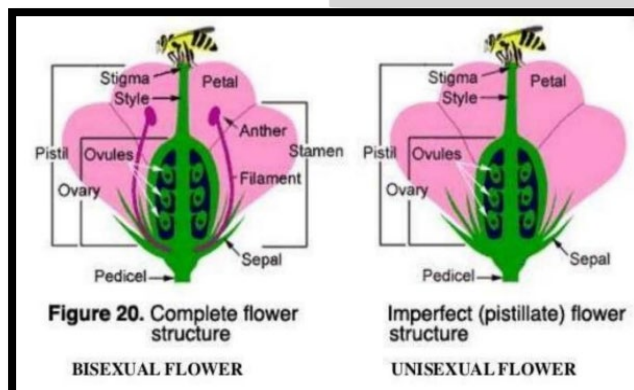
Plant Hormone	
Growth Promoter	Growth Inhibitor
▪ Auxin-Primary growth (length) ▪ Cytokine-Promotes Cell division ▪ Gibberellin-Germination	▪ Ethylene-Ripping of fruit ▪ ABA-Promotes dormancy. ▪ Absciscic acid Harmony promotes degeneracy in Plant.



Type of Asexual reproduction	Description	Examples
Vegetative Propagation	New plants grow from existing vegetative parts like roots, stems or leaves.	Potatoes (stems/tubers), sweet potatoes (roots), Bryophyllum (leaves), roses (stem cuttings), onions/garlic (bulbs), strawberries (runners).
Budding	A small, bulb-like projection (bud) forms on the parent cell, grows, and detaches to form a new individual.	Yeast (fungi).
Fragmentation	The parent body breaks into two or more fragments, each capable of growing into a new, independent plant.	Algae like Spirogyra.
Spore Formation	The parent plant produces tiny, protective-coated spores that can survive harsh conditions and develop into new plants under favorable conditions.	Ferns, mosses, and fungi.

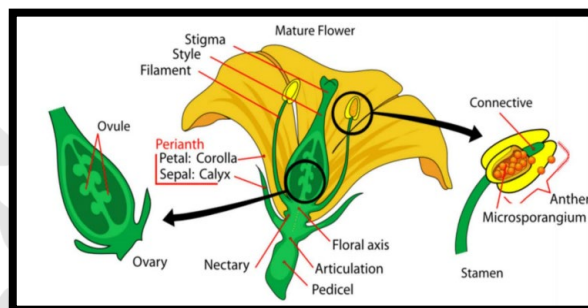
Sexual Reproduction:

- Sexual reproduction in plants typically occurs in flowers and involves the fusion of male and female gametes, which promotes genetic diversity.



Reproductive Parts of a Flower:

- Stamen (male reproductive part) consists of the anther (produces pollen grains, which contain male gametes) and the filament (stalk supporting the anther).
- Pistil or carpel (female reproductive part) consists of the stigma (receives pollen), style (tube connecting stigma to ovary), and ovary (contains ovules, which hold the female gamete or egg cell).



Process:-

- Fertilization:** After landing on the stigma, a pollen tube grows down the style to the ovary. The male gamete travels down this tube and fuses with the female gamete (ovule) in the ovary to form a zygote.
- Seed and Fruit Formation:** The zygote develops into an embryo. The ovule develops into a seed, and the ovary ripens into a fruit, which helps in seed dispersal.

Plant Mineral and Nutrition

- Most of the minerals present in the soil can enter into plants through roots.
- More than 60 types of elements are found in different plants.
- Which elements are found in plants, can be determined by the method called Ash analysis.
- About more than 60 types of elements are present in the plant body but only 17 elements are considered as essential elements.

Macronutrients	Micronutrients
Carbon (C)	Iron (Fe)
Hydrogen (H)	Manganese (Mn)
Oxygen (O)	Boron (B)
Nitrogen (N)	Molybdenum (Mo)
Phosphorus (P)	Copper (Cu)
Calcium (Ca)	Chlorine (Cl)
Potassium (K)	Zinc (Zn)
Magnesium (Mg)	Nickel (Ni)
Sulphur (S)	Cobalt (Co)



- C, H, O, N, K, S, Ca, Fe, Mg, P, Cu, Mn, B, Cl, Zn, Mo, Ni .
- Except these seventeen essential elements there are some beneficial elements such as Na, Si, Se, Co.
- Among 17 essential elements C, H & O are obtained from the air and soil in the form of CO₂ & H₂O not in the form of ions, so these are called nonmineral nutrients or elements.
- Others are mineral nutrients.

Plant mineral and nutrition			
Bacterial Diseases			
Fungal Diseases			
			
Viral Diseases			

Plant Mineral and Nutrition		
Mineral Nutrient	Mobility in Plant	Deficiency Symptoms
Nitrogen (N)	Mobile	Yellowing (chlorosis) of older leaves, stunted growth, poor vigor
Phosphorus (P)	Mobile	Poor root development, delayed flowering, purple or reddish leaves
Potassium (K)	Mobile	Yellowing and scorching of leaf margins, weak stems, lodging
Calcium (Ca)	Immobile	Death of growing tips, leaf tip burn, blossom end rot (tomato)
Magnesium (Mg)	Mobile	Interveneal chlorosis in older leaves leaf curling
Sulphur (S)	Immobile	Uniform yellowing of young leaves, stunted growth
Iron (Fe)	Immobile	Uniform yellowing of young leaves, stunted growth
Manganese (Mn)		Interveneal chlorosis with brown spots

Zinc (zn)	Zinc (Zn)	Zinc (Zn)
Copper (Cu)	Immobile	Dieback of shoots, leaf curling
Boron (B)	Immobile	Death of growing points, brittle leaves, cracked fruits
Molybdenum (Mo)	Mobile	Whiptail in cauliflower, poor nitrogen fixation
Chlorine (Cl)	Mobile	Leaf wilting, bronzing, root stunting

Plant Bacterial Disease

- Plant bacterial diseases cause significant agricultural losses and are characterized by symptoms such as wilting, soft roots, leaf spots, blights, and galls.
- These diseases are caused by major bacterial genera like Pseudomonas, Xanthomonas, Ralstonia, and Erwinia.

Plant Bacterial Disease			
Disease Name	Causal Agent (Bacteria)	Typical Symptoms	Common Hosts
Bacterial Wilt	Ralstonia solanacearum	Rapid wilting, vascular browning, slimy ooze	Tomato, Potato, Tobacco, Banana
Fire Blight	Erwinia amylovora	Scorched appearance, shepherd's crook twigs	Apple, Pear, Rose family
Citrus Canker	Xanthomonas axonopodis pv. citri	Necrotic leaf/fruit spots with yellow halo	Citrus (Lemon, Lime, Orange)
Crown Gall	Agrobacterium tumefaciens	Tumors/galls on roots or stems	Grapevines, Fruit trees, Ornamentals
Black Rot	Xanthomonas campestris pv. campestris	V-shaped yellowing, black veins	Cabbage, Broccoli, Cauliflower
Bacterial Leaf Spot	Pseudomonas syringae	Water-soaked spots, necrosis, blights	Tomato, Beans, Peppers
Soft Rot	Pectobacterium carotovorum	Rapid softening and rotting of tissues	Potato, Carrot, Tomato, Lettuce
Rice Blight	Xanthomonas oryzae pv. oryzae	Leaf blight, yellowing	Rice





		reduced yield	
Olive knot	Pseudomonas savastanoi	Galls/knots on twigs and branches	Olive trees
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Banana Bunchy Top	Banana bunchy top virus (BBTV)	Banana	Bunchy top leaves, severe stunting
Papaya Ringspot	Papaya ringspot virus (PRSV)	Papaya, Cucurbits	Mosaic, ringspots on fruits, blistered leaves
Citrus Tristeza	Citrus tristeza virus (CTV)	Citrus trees	Stem pitting, root decay, plant decline/death

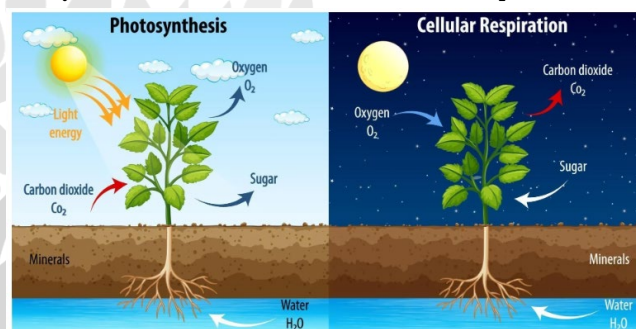
Plant Viral Disease

- Plant viral diseases cause major agricultural losses and are identified by symptoms such as mosaic patterns, leaf curling, chlorosis, necrosis, and stunted growth.
- Unlike bacterial diseases, viruses cannot be controlled directly with chemicals, making prevention and management strategies crucial.

Plant Viral Disease			
Disease Name	Causal Agent (Bacteria)	Common Hosts	Key Symptoms
Tobacco Mosaic	Tobacco mosaic virus (TMV)	Tomato, Tobacco, Pepper	Mosaic patterns, mottling, leaf distortion
Tomato Spotted Wilt	Tomato spotted wilt virus (TSWV)	Tomato, Pepper, Lettuce	Bronze spots, wilted leaves, necrotic ring spots
Tomato Yellow Leaf Curl	Tomato yellow leaf curl virus (TYLCV)	Tomato	Yellowing edges, leaf curling, stunted growth
Cucumber Mosaic	Cucumber mosaic virus (CMV)	Cucumber, Tomato, Banana, Cucumber, Tomato, Banana	Mosaic, mottling, leaf distortion
Potato Y	Potato virus Y (PVY)	Potato, Tomato, Pepper	Vein clearing, crinkling, leaf necrosis

Respiration in plants

- The breaking of C - C single bond of complex compounds through oxidation within the cells, leading to release of considerable amounts of energy and the trapping of this energy for synthesis of ATP is called cellular respiration.



- During cellular respiration or oxidation within a cell, all the energy contained in compounds is not released in a single step because in single if the complete energy is released it, may be converted into heat.
- So the energy during cellular respiration is released in a series of slow step wise reactions controlled by enzymes and it is trapped as chemical energy in the form of ATP.
- ATP thus synthesised, is broken down whenever and wherever energy needs to be utilised, hence ATP act as the energy currency of the cell.
- The compounds that are oxidised during the process of cellular respiration are called respiratory substrates, these are usually carbohydrates but except these fats, proteins & organic acids can also be used as respiratory substrates.
- Primarily carbohydrates (mainly glucose) are used as respiratory substrate.
- In the absence or less availability of carbohydrates, the respiratory substrates can be fats & proteins.





Plant Fungal Disease

Fungal plant diseases are diseases in plants caused by different types of fungi (and fungus-like organisms). These pathogens infect various plant parts such as leaves, stems, roots, and fruits, leading to reduced yield and crop quality.

Disease	Organism	Host Plant	Symptoms
Rust	Puccinia	Wheat, coffee	Reddish-brown pustules on leaves
Smut	Ustilago	Wheat, maize	Black powdery masses
Blight (Late blight)	Phytophthora infestans	Potato, tomato	Dark patches, rotting
Powdery Mildew	Erysiphe	Pea, wheat	White powder on leaves
Downy Mildew	Peronospora	Grapes, onion	Yellow spots, fungal growth below leaf
Ergot	Claviceps purpurea	Rye, wheat	Black elongated sclerotia
Leaf Spot	Alternaria	Many plants	Circular leaf spots
Wilt	Fusarium	Tomato, cotton	Yellowing, wilting
Red Rot	Colletotrichum falcatum	Sugarcane	Red internal stem.
Damping Off	Pythium	Seedlings	Seedling collapse

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